

Unit-IV Tutorial

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1. $\mu=2$ & $\sigma=0.5$; $z = \frac{\ln x - \mu}{\sigma}$

$$\text{Mean} = E(X) = e^{\mu + \frac{\sigma^2}{2}} = e^{2 + \frac{0.5^2}{2}} = 8.3729$$

$$\text{Variance} = e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2} = e^{4 + 2(0.5)^2} - e^{4 + 0.5^2} = 19.9117$$

2. Given, $\text{Mean} = E(X) = 6000 \Rightarrow e^{\mu + \frac{\sigma^2}{2}} = 6000$ — (1)

Also, $\text{Variance} = e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2} = 5000^2$ — (2)

In eq (1),

$$\ln e^{\mu + \frac{\sigma^2}{2}} = 6000 \Rightarrow \mu + \frac{\sigma^2}{2} = 8.6995$$
 — (3)

(3) in (2)

$$\Rightarrow e^{2\mu + \sigma^2} (e^{\sigma^2} - 1) = 5000^2 \Rightarrow e^{2(8.6995)} (e^{\sigma^2} - 1) = 5000^2$$

$$\Rightarrow e^{\sigma^2} = \frac{5000^2}{e^{2(8.6995)}} + 1 \Rightarrow \sigma^2 = \ln(1.6945) = 0.5274$$

$$\Rightarrow \mu = \ln \left[\frac{6000}{e^{0.5274/2}} \right] = 8.4358$$

$$P(X > 8000) = P\left(z > \frac{\ln(8000) - 8.4358}{\sqrt{0.5274}}\right) = 0.2236$$
$$= P(z > 0.76) = 0.5 - 0.2764 = 0.2236$$

3. $n=15$, $p=0.5=q$ $z = \frac{x - np}{\sqrt{npq}} = \frac{x - 7.5}{\sqrt{3.75}}$

$np = 7.5$, $npq = 3.75$

$$P(2 \leq x \leq 7) = P\left(\frac{2 - 7.5}{\sqrt{3.75}} \leq z \leq \frac{7 - 7.5}{\sqrt{3.75}}\right) = P(-2.84 \leq z \leq -0.26)$$
$$= \phi(2.84) - \phi(0.26) = 0.4977 - 0.1026 = 0.3951$$

4. $\mu = \lambda = 45$ $z = \frac{x - \lambda}{\sqrt{\lambda}} = \frac{x - 45}{3\sqrt{5}}$
 $\sqrt{\lambda} = 3\sqrt{5}$

Required prob. = $P(x > 50) = P(z > \frac{5}{3\sqrt{5}}) = P(z > 0.74)$
 $= 0.5 - \phi(0.74) = 0.5 - 0.2704 = 0.2296$

5. Marginal distribution of X: Marginal distribution of Y:

X_i	2	4	6
$f(x_i)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Y_j	1	3	9
$f(y_j)$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{6}$

$E(X) = \sum x_i \cdot f(x_i) = \frac{1}{2} + 2 + \frac{3}{2} = 4$; $E(Y) = \sum y_j \cdot f(y_j) = \frac{1}{2} + 1 + \frac{3}{2} = 3$

$E(XY) = \sum \sum x_i y_j J_{ij} = \frac{2}{8} + \frac{6}{4} + \frac{18}{12} + 1 + \frac{12}{4} + \frac{6}{8} + \frac{18}{24} + \frac{6(9)}{12} = 12$

$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 12 - 12 = 0$

$\text{Corr}(X, Y) = \rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = 0$

6.

$X \backslash Y$	-2	5	8
1	0.21	0.35	0.14
2	0.09	0.15	0.06

$E(X) = \sum x_i \cdot f(x_i)$
 $= 1(0.7) + 2(0.3)$
 $= 1.3$

7. Marginal distribution of X: Marginal distribution of Y:

X_i	1	2
$f(x_i)$	0.6	0.4

Y_j	-2	-1	4	5
$f(y_j)$	0.3	0.3	0.1	0.3

$= 0.6 + 0.3 + 0.1 + 0.3$

i) $E(X) = \sum x_i \cdot f(x_i) = 0.6 + 0.4(2) = 1.4$; $E(Y) = \sum y_j \cdot f(y_j) = -0.6 - 0.3 + 0.4 + 0.5 = 1$

ii) $E(XY) = \sum \sum x_i y_j J_{ij} = -2(0.1) - 0.2 + 5(0.3) - 4(0.2)$
 $- 2(0.1) + 8(0.1) = 0.9$

$$iii) \sigma_x = \sqrt{E(x^2) - [E(x)]^2} = \sqrt{2.2 - 1.4^2} = 0.4899$$

$$\sigma_y = \sqrt{E(y^2) - [E(y)]^2} = \sqrt{10.6 - 1} = 3.0984$$

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$$iv) \text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 0.9 - 1.4 = -0.5$$

$$E(x^2) = \sum x_i^2 f(x_i) = 1/0.6 + 4/0.4 = 2.2$$

$$E(y^2) = 4/0.3 + 0.3 + 16/0.1 + 25/0.3 = 10.6$$

$$v) \rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y} = \frac{-0.5}{0.4899(3.0984)} = -0.3294$$

$$8. i) \Rightarrow \int_0^{50} \int_0^{50} k(x^2 + y^2) dy dx = 1 \Rightarrow k \int_0^{50} \int_0^{50} (x^2 + y^2) dy dx = 1$$

$$\Rightarrow k \int_0^{50} \left[x^2 y + \frac{y^3}{3} \right]_0^{50} dx = 1 \Rightarrow k \int_0^{50} \left(50x^2 + \frac{50^3}{3} - 30x^2 - \frac{30^3}{3} \right) dx = 1$$

$$\Rightarrow k \int_0^{50} \left(20x^2 + \frac{98000}{3} \right) dx = 1 \Rightarrow k \left(\frac{20}{3} x^3 + \frac{98000}{3} x \right) \Big|_0^{50} = 1$$

$$\Rightarrow k = \frac{3}{3920000}$$

$$ii) f(x, y) = \begin{cases} \frac{3}{3920000} (x^2 + y^2), & 30 \leq x \leq 50, 30 \leq y \leq 50 \\ 0, & \text{otherwise} \end{cases}$$

$$P(30 \leq x \leq 40, 30 \leq y \leq 50) = \int_{30}^{40} \int_{30}^{50} k(x^2 + y^2) dy dx = k \int_{30}^{40} \left[x^2 y + \frac{y^3}{3} \right]_{30}^{50} dx$$

$$= k \int_{30}^{40} \left(50x^2 + \frac{50^3}{3} - 30x^2 - \frac{30^3}{3} \right) dx = k \int_{30}^{40} \left(20x^2 + \frac{98000}{3} \right) dx$$

$$= k \left[\frac{20x^3}{3} + \frac{98000x}{3} \right]_{30}^{40} = k \left(\frac{20(40^3)}{3} + \frac{98000(40)}{3} - \frac{20(30^3)}{3} - \frac{98000(30)}{3} \right)$$

$$= k \left(\frac{1720000}{3} \right) = \frac{3}{3920000} \times \frac{1720000}{3} = 0.4388$$

9. i) Marginal distribution of X: Marginal distribution of Y:

X_i 0 1 2

$f(x_i)$ 0.3 0.6 0.1

Y_j 0 1

$f(y_j)$ 0.6 0.4

$$ii) E(XY) = \sum \sum x_i y_j J_{ij} = 0.2$$

$$iv) \text{cov}(X, Y) = E(XY) - E(X)E(Y) = 0.2 - E(X)E(Y)$$

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$$= 0.2 - \sum x_i f(x_i) \cdot \sum y_j f(y_j) = 0.2 - (0.6 + 0.2)(0.4) = 0.2 - 0.4 = -0.2$$

$$iii) \sigma_x = \sqrt{E(X^2) - [E(X)]^2} = \sqrt{\sum x_i^2 f(x_i) - 0.8^2} = \sqrt{1 - 0.8^2} = 0.6$$

$$\sigma_y = \sqrt{E(Y^2) - [E(Y)]^2} = \sqrt{\sum y_j^2 f(y_j) - 0.4^2} = \sqrt{0.4 - 0.4^2} = 0.4899$$

vi) $\text{cov}(X, Y) \neq 0$, hence X & Y aren't independent.

10. i) Marginal distribution of X Marginal distribution of Y

X_i	0	1	2
$f(x_i)$	$4/15$	$2/5$	$1/3$

X_j	-1	0	1
$f(x_j)$	$2/5$	$1/3$	$4/15$

$$ii) P(Y=y/X=2) : P(Y=1/X=2) = \frac{P(Y=1 \cap X=2)}{P(X=2)}$$

$$= \frac{\frac{2}{15}}{\frac{1}{3}} = \frac{2}{5}$$

$$P(Y=0/X=2) = \frac{P(Y=0 \cap X=2)}{P(X=2)} = \frac{\frac{1}{15}}{\frac{1}{3}} = \frac{1}{5}$$

$$P(Y=1/X=2) = \frac{P(Y=1 \cap X=2)}{P(X=2)} = \frac{\frac{2}{15}}{\frac{1}{3}} = \frac{2}{5}$$

$$iii) P(X=x/Y=-1) : P(X=0/Y=-1) = \frac{P(X=0 \cap Y=-1)}{P(Y=-1)} = \frac{\frac{1}{15}}{\frac{2}{5}} = \frac{1}{6}$$

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$$P(X=1 | Y=-1) = \frac{P(X=1 \cap Y=-1)}{P(Y=-1)} = \frac{\frac{1}{15}}{\frac{2}{5}} = \frac{1}{6}$$

$$P(X=2 | Y=-1) = \frac{P(X=2 \cap Y=-1)}{P(Y=-1)} = \frac{\frac{1}{15}}{\frac{2}{5}} = \frac{1}{6}$$